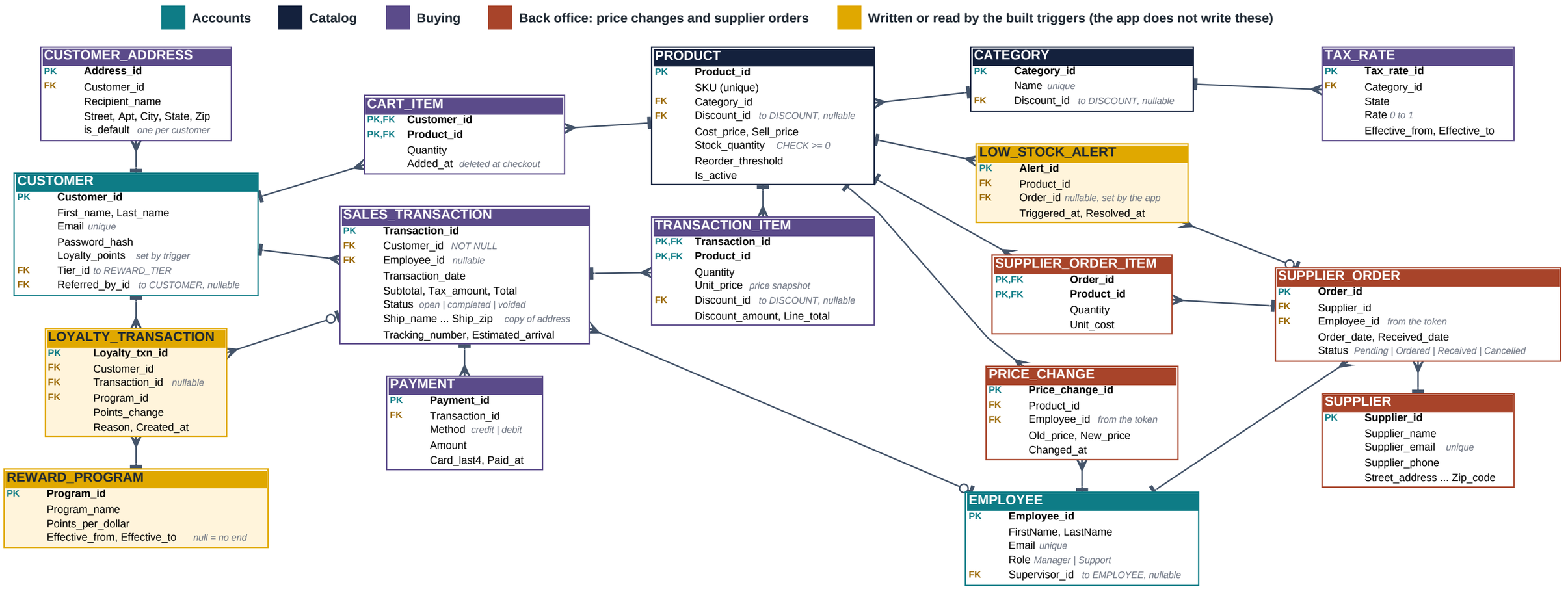
Core tables, price changes and supplier orders

17 of the 27 tables: the 13 core tables for accounts and buying, plus the 4 tables behind manager price changes and ordering products from suppliers. Prepared Oct 21, 2026.



Reading the lines. A bar means one parent row, a crow's foot means many child rows, and a circle means optional (the foreign key can be NULL). Three links are optional: EMPLOYEE to SALES_TRANSACTION (an online order has no employee), SALES_TRANSACTION to LOYALTY_TRANSACTION, and SUPPLIER_ORDER to LOW_STOCK_ALERT (an alert only points to an order once a manager places one).

Price change (manager only). A manager edits `PRODUCT.Sell_price` and the app inserts a PRICE_CHANGE row (old price, new price, the employee id from the login token) in the same query. No trigger does this. The cart and new orders use the new price; finished orders keep the `Unit_price` saved on TRANSACTION_ITEM.

Ordering products from suppliers (manager only; support can only view). 1) A sale drops stock to the reorder threshold and TR_PRODUCT_LowStock opens a LOW_STOCK_ALERT. 2) A manager creates a SUPPLIER_ORDER with SUPPLIER_ORDER_ITEM rows; nothing is ordered automatically. 3) The app links the alert to the order by setting `LOW_STOCK_ALERT.Order_id`. 4) The order moves Pending, Ordered, Received (Cancelled is allowed from Pending or Ordered); items change only while Pending, and the app enforces this. 5) Marking it Received fires TR_SUPPLIER_ORDER_Received, which adds the stock, and TR_PRODUCT_LowStock then closes the alert once stock is above the threshold.

Who is who. CUSTOMER has no role column: a customer signs in against CUSTOMER, staff sign in against EMPLOYEE, and `EMPLOYEE.Role` is Manager or Support. What each role may do is enforced by the Node.js app, not the database. **Minimum data for a first purchase:** a customer with a default address, a category, a product with stock, a TAX_RATE row for that category and the ship-to state, and one REWARD_PROGRAM active on the sale date.

Not drawn (10 tables). DISCOUNT, SALE_EVENT, REWARD_TIER, SUPPLIER_PRODUCT (which supplier sells which product, with cost and lead time), CUSTOMER_PHONE, INVENTORY_ADJUSTMENT, PRODUCT_RETURN, PRODUCT_RETURN_ITEM, REVIEW and NOTIFICATION. Columns that point to them are marked "to ...", and the two self-links (Supervisor_id, Referred_by_id) are not drawn as lines.